

Agent Identity Protocol

AIP v0.1 — White Paper

The Open Standard for AI Agent Identity, Verification, and Governance

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Executive Summary

Autonomous AI agents are executing consequential decisions at enterprise scale — managing workflows, transacting on behalf of organizations, and interacting with sensitive systems — without a standardized mechanism for identity verification, accountability assignment, or governance enforcement. This gap represents a systemic risk that no existing identity standard was designed to address.

The Agent Identity Protocol (AIP) is an open standard that defines how AI agents are credentialed, verified, and governed throughout their operational lifecycle. AIP provides:

- A canonical identity schema for autonomous agents — structured, tamper-evident, and cryptographically signed
- A verification API that enables any system to confirm an agent's identity, authority, and governance status in real time
- A compliance mapping layer that operationalizes obligations under the EU AI Act, Colorado SB 24-205, and NIST AI RMF 1.0
- A revocation mechanism with immediate, irrevocable effect across all verification endpoints

- › *AIP is not a monitoring tool. It is the identity infrastructure layer that makes all other AI governance controls enforceable.*

The reference implementation of AIP is Agent Identity Systems (AIS), operated by CoreIdentity Development Group Inc. AIS provides a production API, developer SDKs, and a governance portal — purpose-built to allow organizations to credential, verify, and audit their agent workforce at scale.

1. The Problem: Ungoverned Agents at Scale

1.1 The Identity Gap

Every regulated interaction in enterprise systems requires identity. A human employee has a badge, a role, an audit trail, and an accountability chain. A software service has a service account, an access policy, and a certificate. An AI agent executing consequential decisions has none of these — unless the organization has built custom governance infrastructure from scratch.

This is not a theoretical concern. The deployment of autonomous agents has outpaced the governance infrastructure required to manage them. Organizations are deploying agents that:

- Execute financial transactions without attribution
- Access sensitive data without auditable authorization records
- Make consequential decisions without disclosing their AI nature to affected parties
- Operate without defined revocation procedures when behavior deviates from policy

1.2 Why Existing Standards Fail

Existing identity standards — OAuth 2.0, SAML, SCIM, OpenID Connect — were designed for human users and static software services. They carry fundamental assumptions that do not hold for autonomous agents:

- Static credentials: Service accounts assume deterministic, policy-bound behavior. Autonomous agents adapt, learn, and deviate.
- Human-in-the-loop authorization: OAuth flows assume a human is present to grant consent. Autonomous agents operate without a consent moment.
- No capability semantics: Existing standards define access permissions but not behavioral capability bounds — the distinction that matters for agents.
- No governance policy attachment: A JWT carries claims about identity. It carries nothing about the governance policy that constrains what that identity may do.

› *Extending existing identity standards to cover autonomous agents is architecturally equivalent to using a driver's license to credential a self-driving vehicle. The instrument was not designed for the entity.*

1.3 The Regulatory Mandate

Regulators have moved. The EU AI Act (effective August 2024) establishes binding obligations for organizations deploying high-risk AI systems, including explicit requirements for risk management systems, transparency mechanisms, and quality management documentation at the individual system level. Colorado SB 24-205 creates developer and deployer obligations for algorithmic transparency and due diligence. NIST AI RMF 1.0 establishes a comprehensive risk management framework that presupposes the ability to identify, trace, and govern individual AI systems.

None of these frameworks can be operationalized without answering a foundational question: which agent did this, under what authority, and who is accountable? AIP is the answer to that question.

2. The Agent Identity Protocol

2.1 Design Principles

AIP is designed around five first-principles constraints:

- **Verifiability:** Any relying party must be able to confirm an agent's identity and governance status without trusting the agent's self-assertion.
- **Portability:** AIP credentials must be interpretable across organizations, platforms, and enforcement systems without proprietary dependencies.
- **Least Privilege:** The default credential grants no capabilities. Every capability must be explicitly declared and policy-justified.
- **Immutability:** Issued credentials are cryptographically sealed. Modifications require revocation and reissuance — preserving audit integrity.
- **Quantum Resistance:** All cryptographic operations use NIST-standardized post-quantum algorithms — ML-KEM-768, ML-DSA-65, SLH-DSA-128s — ensuring credential integrity against both classical and quantum adversaries.

2.2 The AIP Credential Schema

An AIP credential is a structured JSON-LD document, cryptographically signed at issuance and verifiable at any subsequent point. The following table defines the canonical field set for AIP v0.1:

Field	Type	Description
<code>agent_id</code>	<code>string (UUID)</code>	Globally unique agent identifier — primary key for all AIP operations
<code>agent_name</code>	<code>string</code>	Human-readable display name. Required for transparency compliance.
<code>owner_id</code>	<code>string (UUID)</code>	Cryptographically linked to the issuing organization or developer account
<code>credential_hash</code>	<code>string (SHA-256)</code>	Tamper-evident hash of the credential payload at issuance time
<code>issued_at</code>	<code>ISO 8601</code>	UTC timestamp of credential issuance — immutable after signing
<code>expires_at</code>	<code>ISO 8601 null</code>	Null indicates non-expiring credential. Recommended: 90-day rotation cycle
<code>risk_tier</code>	<code>enum</code>	<code>low medium high critical</code> — drives enforcement scope in Sentinel OS

Field	Type	Description
<code>governance_policy</code>	<code>string (URI)</code>	Pointer to the governing policy document at time of issuance
<code>capabilities</code>	<code>string[]</code>	Explicit capability manifest — AIP enforces least-privilege by default
<code>enforcement_scope</code>	<code>object</code>	Data access bounds, action permissions, and geographic constraints
<code>audit_trail</code>	<code>object[]</code>	Append-only log of all verification events. Signed per-entry.
<code>status</code>	<code>enum</code>	active suspended revoked — revocation is immediate and irrevocable
<code>signature</code>	<code>string (ML-DSA)</code>	Post-quantum digital signature over the full credential payload

2.3 The Credential Lifecycle

Issuance

Credentials are issued through an authenticated API call by an authorized owner account. The issuance pipeline enforces:

- Governance policy attachment — a valid policy document URI must be provided
- Capability declaration review — capabilities are validated against the owner's authorized scope
- Risk tier assignment — determines enforcement profile applied by downstream systems
- Cryptographic signing — ML-DSA-65 signature over the complete credential payload

Verification

Any relying party can verify an AIP credential via a single API call to the AIS verification endpoint. The response is a deterministic, signed attestation of:

- Identity validity — the `agent_id` exists and the credential is authentic
- Governance status — the credential is active, not suspended or revoked
- Capability scope — the requested action falls within the agent's declared capability manifest
- Policy currency — the attached governance policy is the current version

Revocation

Revocation is immediate and irrevocable. A single authenticated API call transitions a credential to revoked status. All subsequent verification requests against a revoked credential return a 403 response with a signed revocation attestation. There is no grace period and no partial revocation — the design is intentional. Governance controls must be absolute to be meaningful.

2.4 API Reference — Core Endpoints

Endpoint	Method	Description
/agents	POST	Issue a new AIP credential. Requires authenticated owner account.
/agents/{id}/verify	GET	Verify credential validity, governance status, and capability scope.
/agents/{id}	GET	Retrieve full credential record for authorized owner.
/agents/{id}/revoke	POST	Immediately and irrevocably revoke a credential.
/billing/usage	GET	Retrieve usage metrics and quota status for the current period.

3. Compliance Mapping

AIP is designed to operationalize — not merely acknowledge — regulatory obligations. The following table maps AIP v0.1 capabilities to specific requirements across the three primary governance frameworks applicable to enterprise AI deployment in 2026:

Framework	Requirement	AIP Mapping	Coverage
EU AI Act (2024)	Art. 9 – Risk Mgmt	AIP credential carries risk tier, enforcement scope, and accountability chain per Art. 9 requirements	✓ Full
EU AI Act (2024)	Art. 13 – Transparency	Every agent interaction produces a signed, human-readable audit record disclosing identity and decision rationale	✓ Full
EU AI Act (2024)	Art. 17 – QMS	AIP issuance requires governance policy attachment — operationalizing Quality Management System requirements	✓ Full
CO SB 24-205	§6-1-1702 Disclosure	AIP mandates disclosure of AI identity at point of interaction — satisfying Colorado consumer protection disclosure	✓ Full
CO SB 24-205	§6-1-1703 Due Diligence	Credential issuance pipeline enforces pre-deployment policy review — satisfying developer due diligence obligations	✓ Full
NIST AI RMF 1.0	GOVERN 1.1	AIP anchors organizational AI governance policy to individual agent credentials — traceable to GOVERN function	✓ Full
NIST AI RMF 1.0	MAP 1.5	Risk classification embedded in every credential — operationalizing MAP function context and risk identification	✓ Full
NIST AI RMF 1.0	MEASURE 2.5	Continuous verification logging produces the metrics infrastructure required by MEASURE function	✓ Full
NIST AI RMF 1.0	MANAGE 3.2	Killswitch and revocation capabilities satisfy MANAGE function incident response requirements	✓ Full

› *AIP does not assert compliance on behalf of any organization. It provides the identity and audit infrastructure that makes organizational compliance claims verifiable and defensible.*

4. Reference Implementation: Agent Identity Systems

4.1 Overview

Agent Identity Systems (AIS) is the production reference implementation of AIP v0.1, operated by CoreIdentity Development Group Inc. AIS provides the full AIP credential lifecycle — issuance, verification, revocation, and audit — through a hosted API, developer SDKs, and a governance portal.

- API endpoint: `api.agentidentity.systems`
- Developer portal: `agentidentity.systems`
- TypeScript SDK: `@agentidentity/sdk` (npm)
- Python SDK: `agentidentity` (PyPI)

4.2 Production Metrics

As of April 2026, AIS is operating under continuous soak test conditions targeting 100,000 verified agent calls. The soak infrastructure executes a complete credential issuance, verification, and audit cycle on a 30-minute cadence, producing a tamper-evident Golden Record for each interaction signed against EU AI Act and Colorado SB 24-205 compliance markers.

4.3 Security Architecture

AIS implements the full AIP cryptographic specification:

- ML-KEM-768: Post-quantum key encapsulation for all credential transport
- ML-DSA-65: Post-quantum digital signatures over credential payloads
- SLH-DSA-128s: Stateless hash-based signatures for audit trail entries
- SHA-3-512: Credential hashing and integrity verification
- AES-256-GCM: Symmetric encryption for credential storage

All cryptographic primitives are NIST-standardized post-quantum algorithms. AIS completed post-quantum hardening across its full enforcement stack in Q1 2026 — the first AI governance platform to publish this attestation.

4.4 Integration

Integration with AIS requires three steps:

- Account creation at agentidentity.systems — generates an owner_id and API key
- Agent credential issuance — POST /agents with agent metadata, capability manifest, and governance policy URI
- Verification integration — GET /agents/{id}/verify at any decision boundary where agent authority must be confirmed

SDK integration reduces the verification call to a single function invocation with no raw HTTP handling required.

5. Adoption Path and Governance Tiers

5.1 Who Should Implement AIP

AIP is relevant to any organization that deploys autonomous AI agents in an operational context. Priority adoption segments include:

- **Enterprise:** Organizations subject to AI Act obligations deploying agents in customer-facing or decision-making roles
- **Financial Services:** Institutions deploying agents in trading, credit decisioning, or customer advisory contexts
- **Healthcare:** Systems deploying agents in clinical decision support, prior authorization, or patient communication
- **Legal and Compliance:** Firms deploying agents in contract review, regulatory filing, or due diligence workflows
- **IAM Vendors:** Identity and access management platforms seeking to extend their governance surface to AI agents
- **Sovereign Nations:** Government entities deploying AI in public services requiring full auditability and accountability

5.2 AIS Service Tiers

Tier	Monthly Volume	Pricing	Intended Use
Free	1,000 calls	No cost	Developer evaluation and proof-of-concept
Pro	50,000 calls	Contact AIS	Production deployment — single team or product
Enterprise	Unlimited	Contact AIS	Enterprise deployment — SLA, dedicated support, custom policy

6. AIP Roadmap

AIP v0.2 — Q3 2026

- Federated verification — cross-organizational credential validation without shared infrastructure
- Agent-to-agent delegation chains — governed handoff of authority between agents in multi-agent workflows
- W3C Verifiable Credentials alignment — interoperability with existing DID/VC infrastructure

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- Formal specification publication — standards body submission
- Multi-vendor interoperability testing — open test suite for AIP-compatible implementations
- Sector-specific profiles — healthcare, financial services, and public sector credential profiles with pre-mapped compliance overlays

Conclusion

The deployment of autonomous AI agents without a standardized identity and governance infrastructure is the defining enterprise risk of the current AI adoption cycle. Organizations that deploy agents today without accountability mechanisms are accumulating regulatory, operational, and reputational liability at a rate that existing governance frameworks cannot discharge.

AIP is the infrastructure layer that closes this gap. It does not replace organizational governance policy — it makes policy enforceable at the agent level, in real time, with cryptographic auditability. AIS is available today for evaluation, integration, and production deployment.

› *The question is not whether AI agents will be governed. The question is whether organizations will be in control of that governance — or whether regulators will impose it after the fact.*

Contact and Access

Developer Portal: agentidentity.systems

API Documentation: agentidentity.systems/docs

Contact: coreidentitygroup.com/contact

For enterprise, partnership, and technical inquiries, use the contact form at coreidentitygroup.com/contact. We respond to all submissions as capacity allows.

CoreIdentity Development Group Inc. is a Wyoming C-Corporation operating AI governance infrastructure for enterprise and sovereign deployments. Incorporated 2025.